Eduardo José Barroso, Ph.D.



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Born: 11 June 1995 (Age: 30)

Research Experience

08/2024 — Present — **Postdoctoral Researcher**, LAPP, Annecy, France.
Advisors: Thibault Guillemin, Dominique Boutigny.

10/2022 — 07/2023 — **Ph.D Exchange Researcher**, Laboratoire de Physique Subatomique et de Cosmologie (LPSC), Grenoble, France.
Advisor: Céline Combet.

2020 — 2024 — **Ph.D Researcher**, Universidade Estadual de Londrina (UEL), Londrina, Brazil.
Thesis: Primordial Black Hole formation in contracting universes.
Advisor: Sandro Dias Pinto Vitenti.

Education

2020 – 2024 — **Ph.D in Physics**, Universidade Estadual de Londrina (UEL), Londrina, Brazil.

10/2022 — 07/2023 — **Ph.D Exchange Program**, Laboratoire de Physique Subatomique et de Cosmologie (LPSC), Grenoble, France.

08/2018 — 07/2020 — **Masters in Physics**, Universidade Estadual de Londrina (UEL), Londrina, Brazil.

2013-2018 — **Bachelor in Chemical Engineering**, Universidade Estadual de Maringá (UEM), Maringá, Brazil.

Research Publications

Journal Articles

1. **Barroso, Eduardo J**, L. Demétrio, S. Vitenti, and X. Ye, "Primordial black hole formation in a dust bouncing model," *Journal of Cosmology and Astroparticle Physics*, vol. 2025, no. 01, p. 052, 2025.
2. C. Payerne, Z. Zhang, **Barroso, Eduardo J**, *et al.*, "Analysis of the weak lensing mass-richness relation of redmapper clusters in the lsst desc dc2 simulations," *Accepted for publication at Astronomy and Astrophysics. Preprint arXiv:2502.08444*, 2025.
3. S. D. Vitenti, F. de Simoni, M. Penna-Lima, and **Barroso, Eduardo J**, "Autoknots: Adaptive knot allocation for spline interpolation," *Astronomy and Computing*, p. 100 970, 2025.
4. S. D P Vitenti and **Barroso, Eduardo J**, "Apes: Approximate posterior ensemble sampler," *Monthly Notices of the Royal Astronomical Society*, vol. 525, no. 1, pp. 73–90, 2023.

Skills

Languages	Portuguese (native), English (fluent), French (advanced), German (basic).
Coding	Python, C, LaTeX, MCMC, Data Analysis.
Software	SQL, Git.
Misc.	Academic research, teaching, numerical cosmology.

Teaching

2014 **Calculus Monitor** at Universidade Estadual de Maringa UEM $\sim 60h$

2021 **Teaching Assisting plus Monitor** for Statistical Mechanics at Universidade Estadual de Londrina UEL $\sim 30h$

2022 **Teaching Assisting plus Monitor** for Quantum Mechanics and Data Analysis at Universidade Estadual de Londrina UEL $\sim 30h$

Participation in Research Projects

- LSST Dark Energy Science Collaboration (DESC), Full Member.
- LSST DESC Builder (since 2026), a lifetime status for major contributions to DESC software and community, with authorship on all DESC Key Papers.
- Member of the LSST DESC Cluster Cosmology Working Group.
- Development and validation work for the LSST cluster Pipeline.
- Scientific interactions with multiple DESC working groups relevant to cluster cosmology and survey data analysis.

Oral Talks

2023 LSST France meeting talk, Paris, France

2024 LSST France meeting talk, Paris, France

2025 LSST France meeting talk, Orsay, France

Online Seminar at Observatório Nacional, Rio de Janeiro, Brasil.

Seminar at LAPP, Annecy, France.

Supervision

2025 **Supervision of L3 internship**, Jules Maquinghen "Fitting the scaling relation of Wazp DC2 from LSST"(LAPP, France)

Outreach, Knowledge Dissemination, and Valorisation

2025 Facilitator of the "Rubin First Look" event at LPSC, Grenoble, France.

Participation in the "Rubin First Look" event at LAPP, Annecy, France.

Contributor to open-source scientific software used by the cosmology community, including `NumCosmo`, `CROW`, and `CLMM`.

Outreach, Knowledge Dissemination, and Valorisation (continued)

2025 — Participation in the Journée des Doctorants et Postdocs (Grenoble–Annecy), an inter-laboratory scientific dissemination and exchange event.

Exam Committees

2025 — **Master's qualifying exam committee**, Thais Mikami Ornellas, "Relações concentração-mass em escalas de grupos e aglomerados de galáxias: impacto nas restrições de massa e parâmetros cosmológicos", Universidade Estadual de Londrina, Londrina, Brasil.

Master's exam committee, Thais Mikami Ornellas, "Relações concentração-mass em escalas de grupos e aglomerados de galáxias: impacto nas restrições de massa e parâmetros cosmológicos", Universidade Estadual de Londrina, Londrina, Brasil.

Research Responsibilities

- Development and coordination of the DESC-Cluster analysis pipeline

Scientific Animation Activities

2024–Present — Organizer of the biweekly LSST Group meetings at LAPP, Annecy, France.

2025 — Organizer of the Cluster Pipeline Workshop at LAPP, Annecy, France.

2020–2024 — Co-organizer of the Cosmology Journal Club at Universidade Estadual de Londrina (UEL), Brazil.

Institutional Responsibilities

2025 — Contribute to establish centralized data storage solutions for the LSST Group at LAPP through the MUST computing center.

Community and Open Science Activities

- Active contributor to open-science practices through participation in the development and public release of scientific software used by the cosmology community (public packages: ·NumCosmo, desc-crow, CLMM).